

- 5 (a) Intrinsic semiconductor has a fully filled valence band and a conduction band that is empty at 0 K. No conduction is possible since there is an absence of mobile charges. In semiconductor, the forbidden energy band is not too wide, hence there is an increasing probability of excitation of electrons from the valence band into the conduction band as the temperature increases.

As the temperature increases, the electrons from the valence band have a higher probability to move into the conduction band. These leave holes in the valence band, which behaves like positive charges. The excited electrons in the conduction band do not fill up the conduction band, merely occupying a small portion of it. When a potential difference is applied, electrons are able to move within the conduction band, contributing to flow of current. The holes in the valence band also allow electrons in the valence band to move into them.

Hence, the increase in the amount of charge carriers as the temperature increases, decreases the resistance of the semiconductor.

- (b) Metal has no forbidden band and the conduction is partially filled with electrons. When a potential difference is applied to the metal, the electrons are able to move within the conduction band easily. An increase in temperature does not significantly increase the number of electrons in the conduction band (no change in the amount of charge carriers).

However, the lattice vibrations increase with an increase in temperature. This impedes the flow of electrons in the material and hence increases the resistance.